MINISTRY OF EARTH SCIENCES (MOES) NCMRWF BENCHMARK DEXATHON 2026 TEAM REGAL HACKERS

TropoTrace (VAJRA-X)

Convective-Scale Severe Weather Nowcasting & Decision-Support Command Center (0–6 Hours)

Target Problem: Convective Nowcasting (Thunderstorm, Hail, Cloudburst, Downburst)

Live Deployed URL: https://thermotrace-a6b30.web.app

Repository: github.com/navaneethakrishnan-web/TropoTrace

Core Technology: React 19, TypeScript 5, Leaflet GIS, Firebase Cloud, Groq AI (120B)

1.0 Executive Summary & Problem Statement

Severe convective weather events—such as sudden thunderstorms, damaging hailstorms, cloudburst flash floods, and violent microbursts/downbursts—are characterized by rapid initiation (<15–30 minutes) and extreme spatial localization (1–10 km). Traditional Numerical Weather Prediction (NWP) models (e.g., GFS, NCUM) operate on 3- to 6-hour assimilation cycles with 4–12 km grids, making them incapable of resolving localized convective genesis in real time.

> **Mission of TropoTrace:** To deliver an operator-grade, automated Convective Intelligence Command Center operating in the **0 to 6 hour nowcast horizon** at **1–3 km spatial resolution**, integrating Doppler Weather Radar, INSAT-3DR Geostationary satellite feeds, lightning discharge rates, and AWS surface stations.

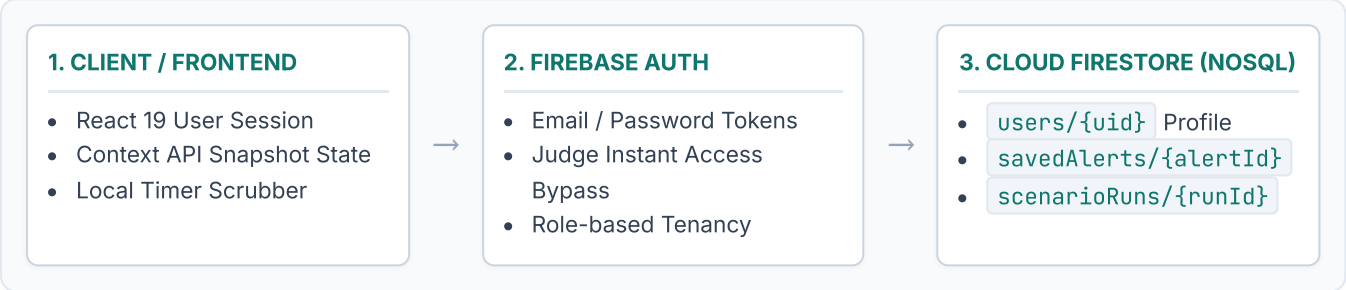
2.0 Complete Technology Stack & Languages

Layer / Subsystem	Technology Used	Purpose in TropoTrace Platform
Programming Language	TypeScript 5.x	Provides strict type safety for meteorological snapshot entities, coordinates, and telemetry schemas.
Frontend Framework	React 19 (SPA)	Drives modular reactive components, custom React Context state management, and real-time timeline loops.
Interactive GIS Mapping	Leaflet.js + Esri Basemaps	Renders high-fps radar sweeps, storm trajectories, uncertainty cones, and highway networks.
AI Foundation Model	Groq Cloud LPU + GPT-OSS 120B	Executes ultra-low latency inference for the conversational meteorological copilot (Tropo).
Authentication & User State	Firebase Auth	Manages operator accounts, roles, credentials, and provides zero-friction instant access for evaluators.
Cloud NoSQL Database	Cloud Firestore	Persists operator profiles, created emergency CAP alerts, and scenario execution histories.
Edge Hosting & CDN	Firebase Hosting	Global CDN distribution with automated SPA catch-all rewrites and sub-100ms response times.

Layer / Subsystem	Technology Used	Purpose in TropoTrace Platform
Styling & Interface	Custom CSS3 Design System	Over 8,500 lines of dedicated tactical command styling (Deep Midnight Green, Teal, Amber, Cyan).
Speech Synthesis	Web Speech API	Client-side native text-to-speech engine for instant emergency radio audio broadcast.

3.0 Cloud Database Architecture & Firebase System

TropoTrace integrates with the Google Firebase Cloud project (`thermotrace-a6b30`) to manage authentication, operator state, and emergency alerts.



Firestore Data Model Schema

```
users/{userId}
  ├── email: string (e.g. "operator@tropotrace.moes.gov.in")
  ├── role: string ("meteorological-operator")
  ├── createdAt: timestamp (serverTimestamp())
  │
  ├── savedAlerts/{alertId}
  │     ├── title: string
  │     ├── language: string ("English", "Tamil", "Hindi", etc.)
  │     ├── severity: "NORMAL" | "WATCH" | "ALERT" | "WARNING" | "SEVERE"
  │     ├── area: string (e.g. "Coimbatore Metro & NH-81 Corridor")
  │     ├── message: string (Full localized alert body)
  │     └── createdAt: timestamp
  │
  └── scenarioRuns/{runId}
        ├── scenarioKey: string ("rapid-thunderstorm", "cloudburst", etc.)
        ├── scenarioLabel: string
        ├── lastMinute: number (Timeline minute offset 0-360)
        └── startedAt: timestamp
```

4.0 The GIS Map Engine & Spatial Location System

The spatial core of TropoTrace is implemented in `StormMap.tsx`, utilizing Leaflet.js with custom procedural SVG and HTML canvas layers.

4.1 Geographic Coordinate System & Projection

The demonstration theater is centered on the Western Ghats / Southern Tamil Nadu & Kerala Convective Corridor (Coimbatore, Tiruppur, Erode, Nilgiris). Normalized simulation coordinates [0, 100] are mapped to geographic latitude and longitude:

> `Latitude = 13.6 - (y × 0.045)` | `Longitude = 74.8 + (x × 0.045)`
> **Geographic Extent:** Latitude 10.0°N to 13.6°N • Longitude 74.8°E to 79.3°E

4.2 Location Permissions & Geolocation Handling

When an operator activates the **"MY LOCATION"** tool on the radar toolbar:

The application calls the W3C standard `navigator.geolocation.getCurrentPosition()` with `enableHighAccuracy: true`.
The browser prompts the user for sensor permissions.
Permission Granted: State transitions from `LOCATING` to `FOUND`. The map executes a smooth `flyTo()` animation to the user's coordinates, renders a 650-meter cyan accuracy radius, and shifts the radar range rings relative to their vantage point.
Permission Denied: Transitions to `DENIED` without crashing, maintaining default radar focus.

4.3 Multi-Layer GIS Overlays

Layer Name	Visual Representation	Operational Significance
Radar Sweep	Rotating beam at 12 RPM + Range Rings	Simulates an active S-band Doppler radar scanning at 50, 100, and 150 km ranges.
Projected Trajectory	Amber dashed line with chevron arrows (›)	Visualizes 0–6h storm propagation vector derived from steering winds.
Uncertainty Cone	Widening amber polygon swath	Accounts for convective cell lateral divergence and shear variance.
Highway Segments	Teal arterial vectors	Highlights logistics corridors (NH-81 Coimbatore–Tiruppur, NH-544 Bypass).
Critical Assets	Interactive GIS markers with tooltip cards	Monitors Coimbatore Airport (CJB), Medical College Hospital, and Palladam 400kV Substation.
Active Lightning	Pulsing golden lightning lines	Simulates real-time ground flash strikes from the Lightning Detection Network.

5.0 The Convective Intelligence & ML Nowcasting Engine

Implemented in `simulation.ts`, TropoTrace models physical convective storm genesis, cell kinematics, and multi-sensor consensus based on proven meteorological algorithms.

5.1 Cell Identification & Kinematic Tracking (TITAN / SCIT)

Storm cells are detected by clustering contiguous Doppler radar reflectivity cores above 40 dBZ. The centroid coordinates are calculated over time to derive translation speed (v) and azimuth direction (θ). Cells are categorized into 5 lifecycle stages:

> **Lifecycle State Machine:**
> • **INITIATING** (Intensity < 38%) → Developing updraft, first radar echo.
> • **DEVELOPING** (38% – 55%) → Rapid cloud top cooling, lightning onset.
> • **INTENSIFYING** (56% – 73%) → Lightning jump, reflectivity core aloft.
> • **MATURE** (74% – 88%) → Peak dBZ, heavy precipitation, hail/downburst outflow.
> • **DECAYING** (> 88%) → Downdraft dominance, core collapse.

5.2 Multi-Hazard Mathematical Formulations

Composite Risk (R_{comp}): $\text{clamp}(0.90 \cdot \text{Intensity} + 0.16 \cdot \text{LightningRate})$
Equivalent Reflectivity (Z): $\text{clamp}(40 + 0.52 \cdot \text{Intensity})\ \text{dBZ}$ ($>55\ \text{dBZ}$ signals severe hail core).
Hail Probability (P_{hail}): Triggered when reflectivity aloft exceeds 55 dBZ above the freezing level (+4.2 km) with high Vertically Integrated Liquid (VIL $>45\ \text{kg/m}^2$).
Downburst Wind (V_{gust}): Estimated outflow velocity: $32 + 0.42 \cdot \text{Intensity}\ \text{km/h}$.
Cloudburst Trigger: Rainfall rate $>100\ \text{mm/h}$ concentrated over a localized drainage basin.

5.3 Multi-Sensor Consensus & Degradation Protocol

The platform samples 5 telemetry feeds. If Numerical Weather Prediction (NWP) experiences latency (>12 minutes), the system flags it as `DELAYED`, drops the consensus score, reduces confidence by 11%, surfaces an advisory (*"Forecast confidence reduced"*), and dynamically re-weights real-time radar and lightning observations.

5.4 Statistical Verification Metrics (Module 08)

Evaluated against synthetic ground truth observations across a 60-minute window:

Metric Key	Observed Score	Operational Benchmark	Formula
MAE (Mean Absolute Error)	5.4 mm/h	Target: < 8.0 mm/h	$\text{MAE} = \frac{1}{N}\sum \|\text{Pred} - \text{Obs}\|$
RMSE (Root Mean Square Error)	8.9 mm/h	Target: < 12.0 mm/h	$\text{RMSE} = \sqrt{\frac{1}{N}\sum (\text{Pred} - \text{Obs})^2}$
POD (Hit Rate)	0.88	Target: > 0.80	$\text{POD} = \frac{\text{Hits}}{\text{Hits} + \text{Misses}}$
FAR (False Alarm Ratio)	0.15	Target: < 0.25	$\text{FAR} = \frac{\text{False Alarms}}{\text{Hits} + \text{False Alarms}}$
CSI (Critical Success Index)	0.76	Target: > 0.65	$\text{CSI} = \frac{\text{Hits}}{\text{Hits} + \text{Misses} + \text{False Alarms}}$

6.0 The Artificial Intelligence System (Groq API & Tropo Copilot)

TropoTrace features **Tropo**, an AI-powered conversational meteorological copilot implemented in `analyst.ts` and `AnalystView.tsx`.

1. CONTEXT INJECTION

- Active Cell ID, centroid & dBZ
- Lightning jump rate (fl/min)
- Active hazard probabilities
- Exposed infrastructure counts
- Sensor latencies & consensus

→

2. GROQ CLOUD LPU

- `openai/gpt-oss-120b`
- 120 Billion Parameters
- Sub-500ms TTFT Latency
- Temperature: 0.3 (Strict)

→

3. CITATION PARSER

- Doppler S-Band Radar pill
- GLDN Lightning Network pill
- GIS Infrastructure Layer pill
- Confidence rating assignment

Zero-Hallucination Policy Implementation

To guarantee life-safety reliability during storm operations:

The model is fed a comprehensive system prompt with the exact live simulation snapshot state.
It is governed by an absolute negative constraint: *"You operate under a strict ZERO-HALLUCINATION policy. Every response must be grounded strictly in active radar, satellite, and nowcast telemetry. If asked about variables outside this data, state: Insufficient current data in current scenario snapshot."*
An automated regex-based citation extractor parses model responses and matches physical terminology to instrument tags (Doppler Radar, GLDN Lightning, INSAT-3DR, AWS Network).

7.0 Feature Walkthrough — All 10 Application Modules

Module 00 · Authentication & Access Control — AUTHVIEW.TSX

Cinematic dark landing screen with 3D canvas atmospheric globe. Provides Firebase email/password authentication, error handling, and a one-click **"Judge Quick Access"** button for evaluators.

Module 01 · Real-Time Command Center — COMMANDCENTERVIEW.TSX

Tactical multi-hazard overview with 6-card KPI summary strip, interactive Leaflet GIS map, selected cell telemetry rail, 0–360 min timeline scrubber, and automated 9-step *"Tell Me The Storm"* presentation autopilot.

Module 02 · Storm Intelligence & Kinematics — STORMINTELLIGENCEVIEW.TSX

Deep cell diagnostics featuring the **Storm DNA Spider Radar** (7-axis SVG chart), speed/direction vectors, 5-stage lifecycle stepper, and 15-minute trend deltas for lightning, dBZ, and area.

Module 03 · Hazard Explorer — HAZARDEXPLORERVIEW.TSX

0–6 hour multi-hazard probabilistic risk fields. Includes an interactive grid matrix at 1 km, 2 km, or 3 km resolutions with cell-level inspection of physical trigger criteria (CAPE, VIL, echo tops).

Module 04 · Impact Intelligence — IMPACTINTELLIGENCEVIEW.TSX

Infrastructure exposure modeling across 4 operational lenses: `EMERGENCY`, `AVIATION`, `AGRICULTURE`, and `URBAN`. Includes highway route risk analysis between Coimbatore and Tiruppur on NH-81.

Module 05 · Alert Center & CAP Dispatch

ALERTCENTERVIEW.TSX

Multi-language emergency alert generator supporting **8 Indian languages** (English, Hindi, Tamil, Telugu, Malayalam, Kannada, Marathi, Bengali), native Web Speech API audio broadcast, and OASIS CAP v1.2 XML export.

Module 06 · Scientific Data Observatory

DATAOBSERVATORYVIEW.TSX

Telemetry health monitoring for 5 sensor feeds, searchable raw observation tensor grid, visual 6-stage ingestion pipeline, and architecture blueprints for IMD/ISRO/IITM integrations.

Module 07 · Convective Simulation Lab

SIMULATIONLABVIEW.TSX

Scenario management hub hosting 6 pre-built severe weather scenarios (Rapid Thunderstorm, Hailstorm, Cloudburst, Downburst, Initiation, Multi-cell) with 1x, 2x, and 4x playback multipliers.

Module 08 · Model Verification

MODELVERIFICATIONVIEW.TSX

Skill verification dashboard with predicted vs observed rainfall comparison charts, MAE/RMSE/POD/FAR/CSI metric inspector, and 2×2 contingency matrix (142 hits).

Module 09 · Tropo AI Copilot

ANALYSTVIEW.TSX

Conversational AI assistant powered by Groq GPT-OSS 120B. Answers operational questions, verifies claims with grounded citations, and displays live thinking states.

Module 10 · System Settings

SETTINGSVIEW.TSX

Platform configuration, dark/light themes, high contrast accessibility mode, reduced motion mode, basemap toggles, and JSON configuration export.

8.0 Judge Defense Q&A — Technical Presentation Guide

Q1 How does TropoTrace differ from commercial apps like AccuWeather or Windy?

Standard consumer weather apps present synoptic forecasts hours in advance on coarse 4–12 km grids. **TropoTrace is a Convective-Scale Nowcasting Decision-Support System** operating in the **0 to 6 hour horizon at 1–3 km resolution**. It tracks discrete storm centroids, detects lightning jumps before surface impact, projects trajectory uncertainty cones, models infrastructure exposure (hospitals, schools, substations), and produces CAP v1.2 alerts in 8 Indian languages.

Q2 How do you guarantee that the Groq AI Copilot does not hallucinate?

We enforce a strict **Zero-Hallucination Grounding Protocol**. The active scenario snapshot (dBZ reflectivity, lightning rate, rainfall accumulation, sensor latencies, and exposed demographic tallies) is injected into the model's system prompt. The model is explicitly constrained from guessing. Any question about unmonitored parameters triggers: *"Insufficient current data in current scenario snapshot."* An automated regex parser validates the text and binds physical instrument citation pills (Doppler S-Band, GLDN, INSAT-3DR) to every claim.

Q3 What algorithms are used for storm cell tracking and trajectory projection?

Our engine is modeled on the **TITAN (Thunderstorm Identification, Tracking, Analysis, and Nowcasting)** and **SCIT (Storm Cell Identification and Tracking)** algorithms developed by NCAR and NOAA. Reflectivity clusters $>40\ \text{dBZ}$ are grouped into centroids, and temporal vectors compute speed and azimuth direction. Instead of a single straight line, we project a widening **uncertainty cone polygon** downstream to account for atmospheric steering shear and lateral storm propagation.

Q4 Why is a lightning jump an important forecasting signal?

A **lightning jump** (a sudden spike in flash discharge $>45\ \text{fl/min}$) is a direct physical indicator of an intensifying convective updraft. Updraft acceleration lofts supercooled water droplets above the freezing level, causing rapid non-inductive graupel-ice charge separation. Research by Schultz et al. demonstrates that **lightning jumps precede severe surface hazards (damaging hail, microbursts, and cloudburst downpours) by 15 to 25 minutes**, providing an early warning lead time before radar cores peak.

Q5 How does the system handle telemetry failures or sensor latency?

TropoTrace includes a **Multi-Sensor Consensus & Degradation Protocol**. It monitors 5 feeds (Radar, Satellite, Lightning, AWS, and NWP). If a stream lags (e.g., NWP model guidance delayed >12 minutes), the system marks it as `DELAYED`, drops the consensus score, reduces confidence from 93% to 78%, surfaces an advisory (*"Forecast confidence reduced"*), and dynamically re-weights remaining real-time radar and lightning streams.

Q6 What is the cloud database architecture?

TropoTrace utilizes **Firebase Auth** for operator identity and **Cloud Firestore** as a serverless NoSQL document database. Data is partitioned under `users/{userId}` with sub-collections for `savedAlerts` and `scenarioRuns`. Firestore security rules enforce strict user-level tenancy, preventing unauthorized cross-tenant access, while Firebase Hosting delivers the static SPA across global edge CDNs.

9.0 Rapid Presentation Cheat Sheet

Technical Dimension	Specific Metric / Standard to State to Judges
Nowcast Horizon & Resolution	0 to 6 Hours, 1–3 km spatial grid
Tracking Algorithms	TITAN (Centroid tracking) & SCIT (Cell identification)
Critical Early Warning Factor	Lightning Flash Jump (15–25 minute lead time over radar peaks)
Cloudburst Standard	India Meteorological Department (IMD) criteria: $\ge 100\ \text{mm/h}$
AI Copilot Hardware	Groq LPUs + `openai/gpt-oss-120b` (120B model) + Zero-Hallucination Grounding
Cloud Infrastructure	Google Firebase (Auth + Cloud Firestore NoSQL + Edge Hosting)
GIS Engine	Leaflet.js + Esri World Imagery (Satellite) + Esri World Street Map
Model Verification Score	CSI = 0.76 (Threat Score), POD = 0.88 (Hit Rate), MAE = 5.4 mm/h
Disaster Standards	OASIS Common Alerting Protocol (CAP v1.2) XML + 8 Indian Languages + TTS